Chapter 5
The Expenditure Cycle Part 1: Purchases and Cash Disbursements Procedures



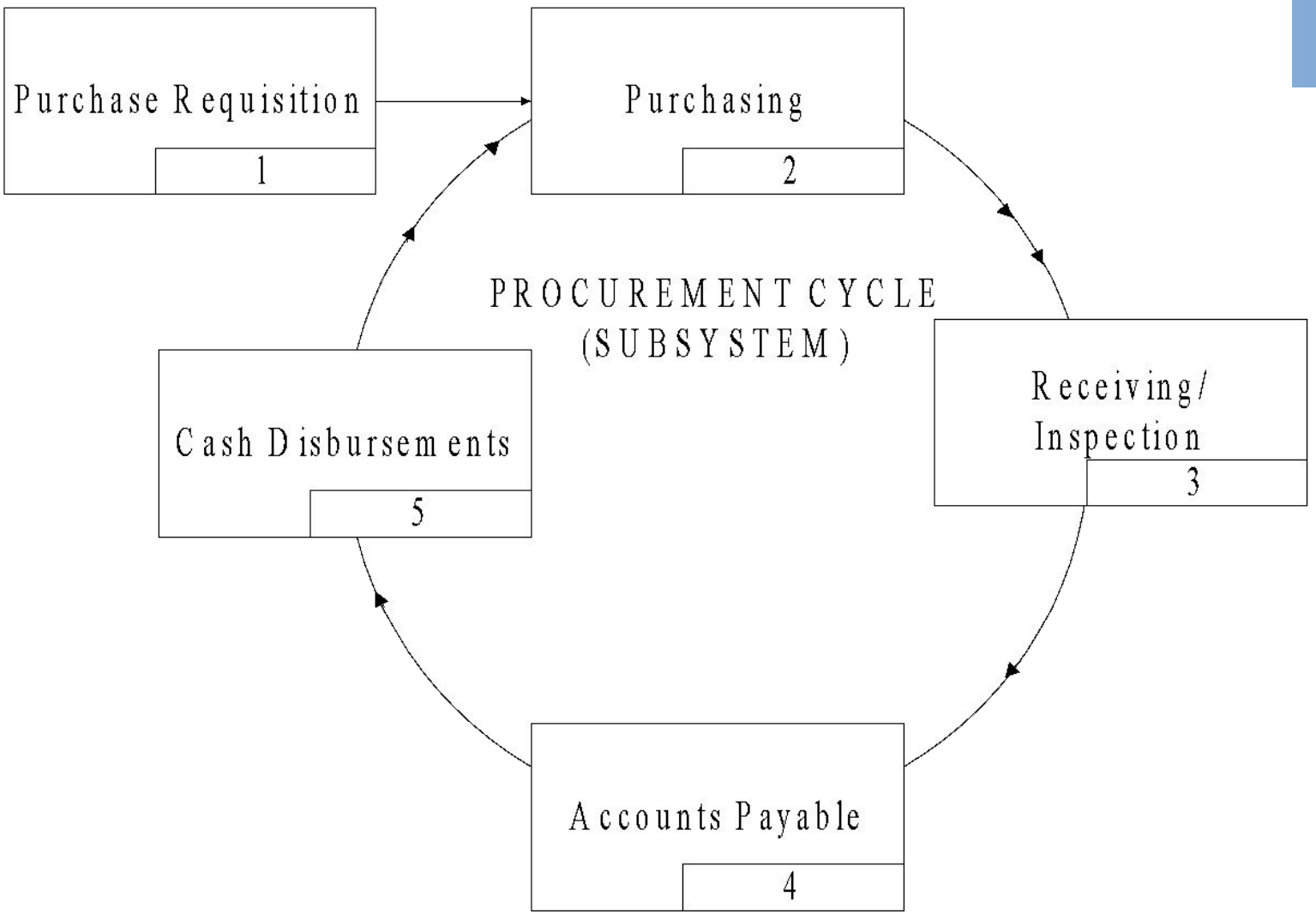
Accounting Information Systems, 7e
James A. Hall

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Objectives for Chapter 5

- Fundamental tasks performed during purchases and cash disbursement processes
- Functional areas involved in purchases and cash disbursements and the flow of these transactions through the organization
- Documents, journals, and accounts that provide audit trails, promote the maintenance of records, and support decision making and financial reporting
- Risks associated with purchase and cash disbursements activities and the controls that reduce these risks
- Operational features and the control implications of technology used in purchases and cash disbursement systems

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Goals of the Expenditure Cycle

- The goal of providing needed resources to organization can be broken down into several objectives:
  - purchase from reliable vendors
  - purchase high quality items
  - obtain best possible price
  - purchase only items that are properly authorized
  - have resources available when they are needed
  - receive only those items ordered
  - ensure items are not lost, stolen, or broken
  - pay for the items in a timely manner

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DFD for Purchase System

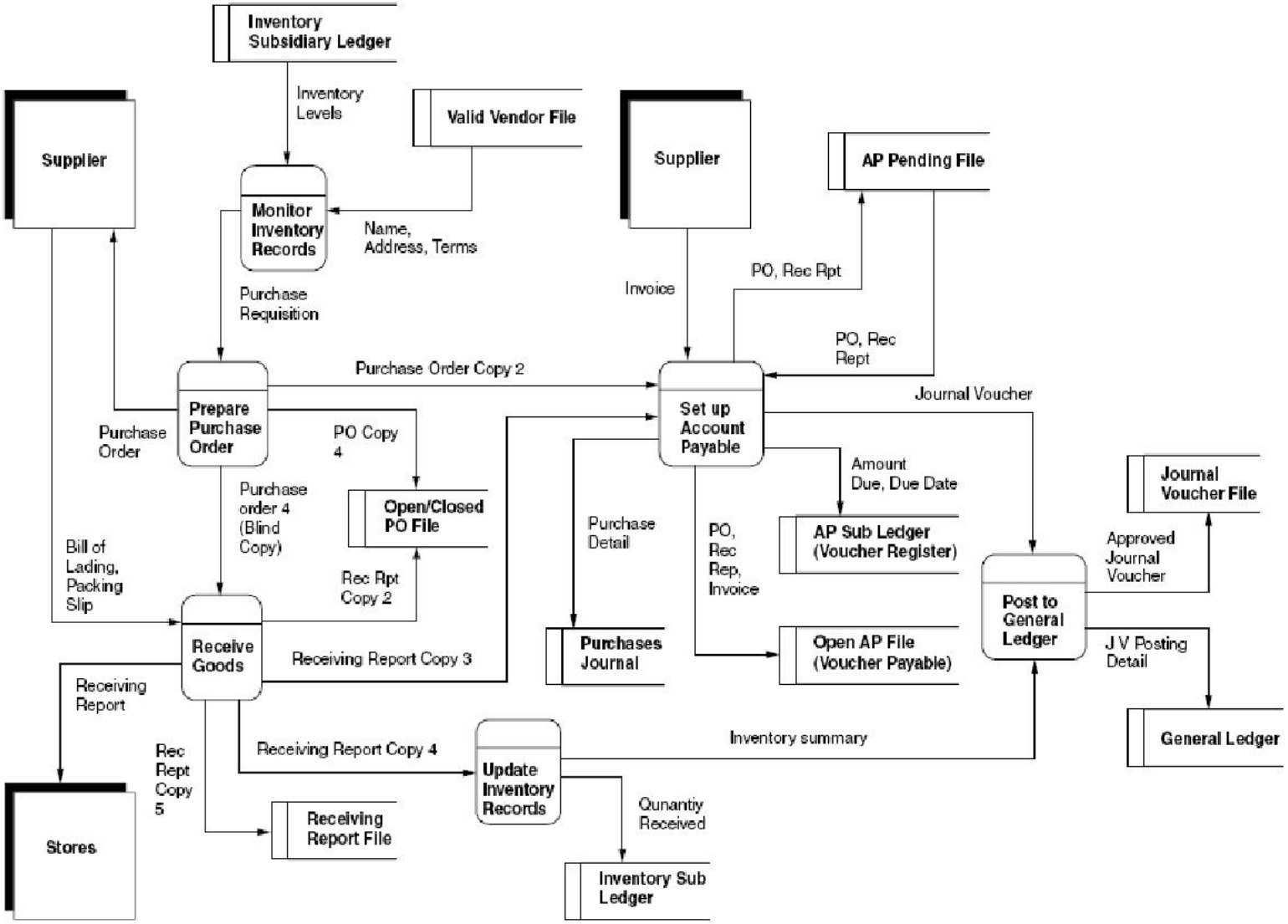


Figure 5-1

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A Purchase System

- Begins in Inventory Control when inventory levels drop to reorder levels
- A purchase requisition (PR) is prepared and copies to sent to Purchasing and Accounts Payable (A/P)
- Purchasing prepares a purchase order (PO) for each vendor and sends copies to Inventory Control, A/P, and Receiving

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A Purchase System

- Upon receipt, Receiving counts and inspects the goods.
  - A blind copy of the PO is used to force workers to count the goods.
- A receiving report is prepared and copies sent to the raw materials storeroom, Purchasing, Inventory Control, and A/P.

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A Purchases System

- A/P eventually receives copies of the PR, PO, receiving report, and the supplier's invoice.
- A/P reconciles these documents, posts to the purchases journal, and records the liability in the accounts payable subsidiary ledger.

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A Purchase System

- A/P periodically summarizes the entries in the purchases journal as a journal voucher which is sent to the General Ledger (G/L) department.

Inv-Control or Purchases DR

Accts Payable-Control CR

- A/P also prepares a cash disbursements voucher and posts it in the voucher register.

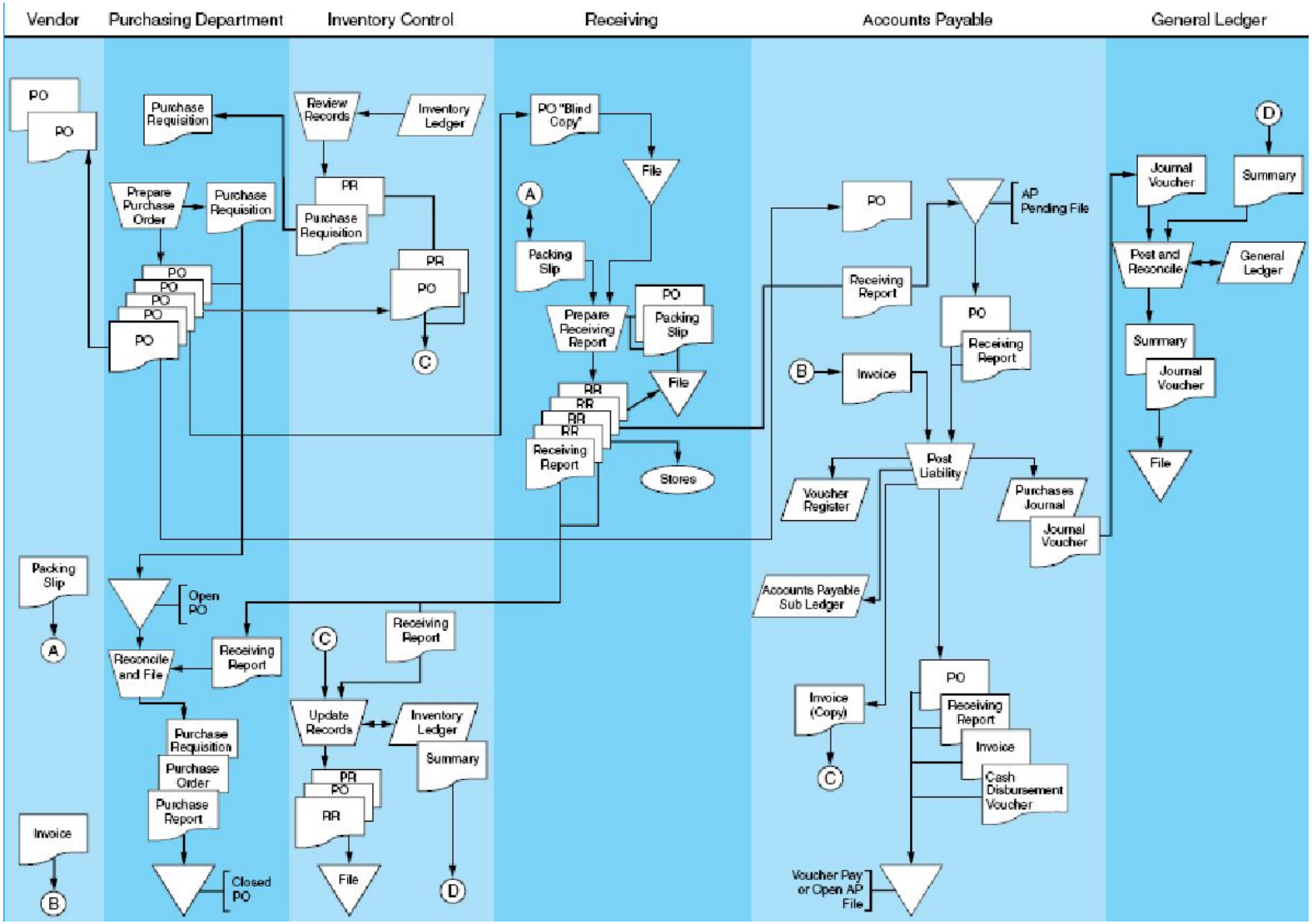
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A Purchase System

- G/L department:
  - posts from the accounts payable journal voucher to the general ledger
  - reconciles the inventory amount with the account summary received from inventory control

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Manual Purchase System



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DFD for Cash Disbursements System

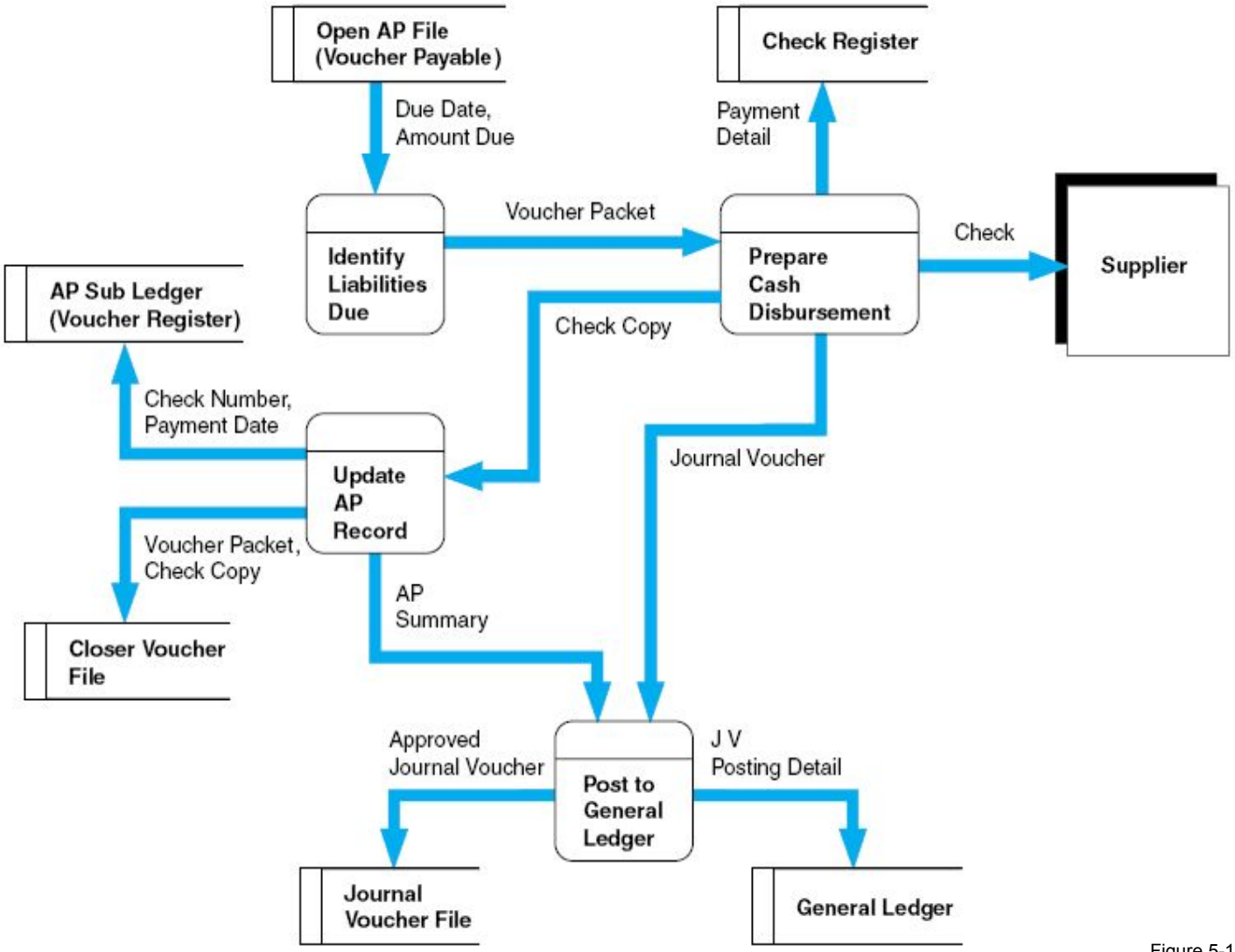


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Cash Disbursements System

- Periodically, A/P searches the open vouchers payable file for items with payments due:
  - A/P sends the voucher and supporting documents to Cash Disbursements
  - A/P updates the accounts payable subsidiary ledger

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Cash Disbursements System

- Cash Disbursements:
  - prepares the check
  - records the information in a check register (cash disbursements journal)
  - returns paid vouchers to accounts payable, mails the check to the supplier
  - sends a journal voucher to G/L:

Accounts Payable ***DR***

Cash ***CR***

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Cash Disbursements System

- G/L department receives:
  - the journal voucher from cash disbursements
  - a summary of the accounts payable subsidiary ledger from A/P
- The journal voucher is used to update the general ledger.
- The accounts payable control account is reconciled with the subsidiary summary.

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Cash Disbursements System

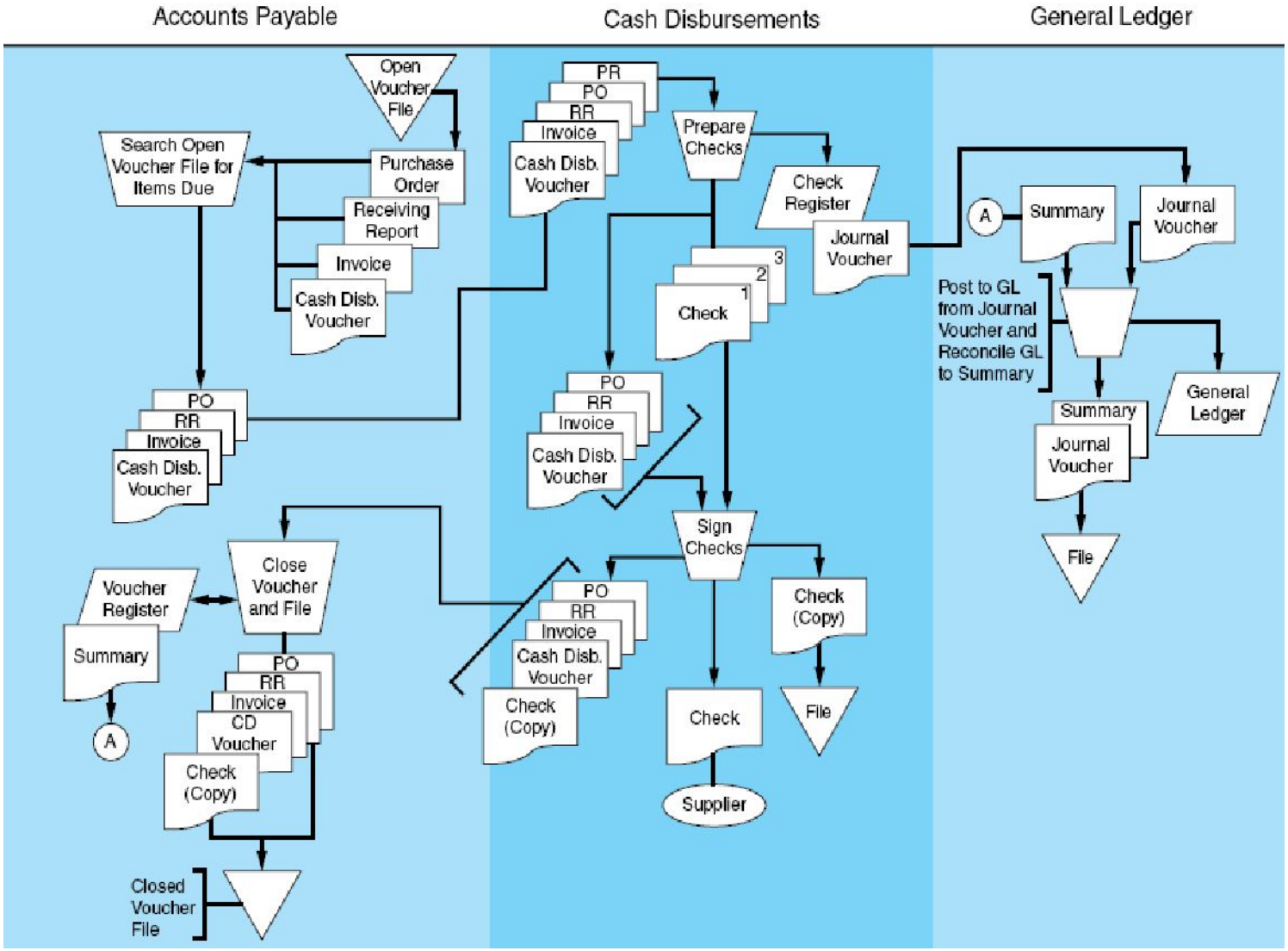


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Computer-Based Accounting Systems

- CBAS technology can be viewed as a continuum with two extremes:
  - **automation** - use technology to improve efficiency and effectiveness
  - **reengineering** – use technology to restructure business processes and firm organization

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Levels of Automating and Reengineering Ordering

- Computer generates PR
  - Purchases manually generates PO
- Computer generates PO (no PR needed)
  - PO not sent until manually reviewed
- Computer-generated PO is automatically sent without manual review
- Electronic Data Interchange (EDI)
  - Computer-to-computer communication without PO

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Expenditure Cycle Database

- Master Files
  - supplier (vendor) master file
  - accounts payable master file
  - merchandise inventory master file
- Transaction and Open Document Files
  - purchase order file
    - open purchase order file
  - supplier's invoice file
  - open vouchers file
  - cash disbursements file
- Other Files
  - supplier reference and history file
  - buyer file
  - accounts payable detail file

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Computer-Based Purchases

- A Data Processing dept. performs routine accounting tasks.
- Purchasing - a computer program identifies inventory requirements
- The following methods are used for authorizing and ordering inventories:
  - the system prepares POs and sends them to Purchases for review, signing, and distributing
  - the system distributes POs directly to the vendors and internal users, bypassing Purchases
  - the system uses electronic data interchange (EDI) and electronically places the order without POs

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Computer-Based Purchases

- Other tasks performed automatically by the computer:
  - updates the inventory subsidiary file from the receiving report
  - calculates batch totals for general ledger update
  - closes the corresponding records in the open PO file to the closed PO file
  - validates the voucher records against valid vendor files

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Computer-Based Cash Disbursements

- Tasks performed automatically by the computer:
  - the system scans for vouchers currently due
  - prints checks for these vouchers
  - records these checks in the check register
  - batch totals are prepared for the general ledger update procedure

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Batch Purchases System

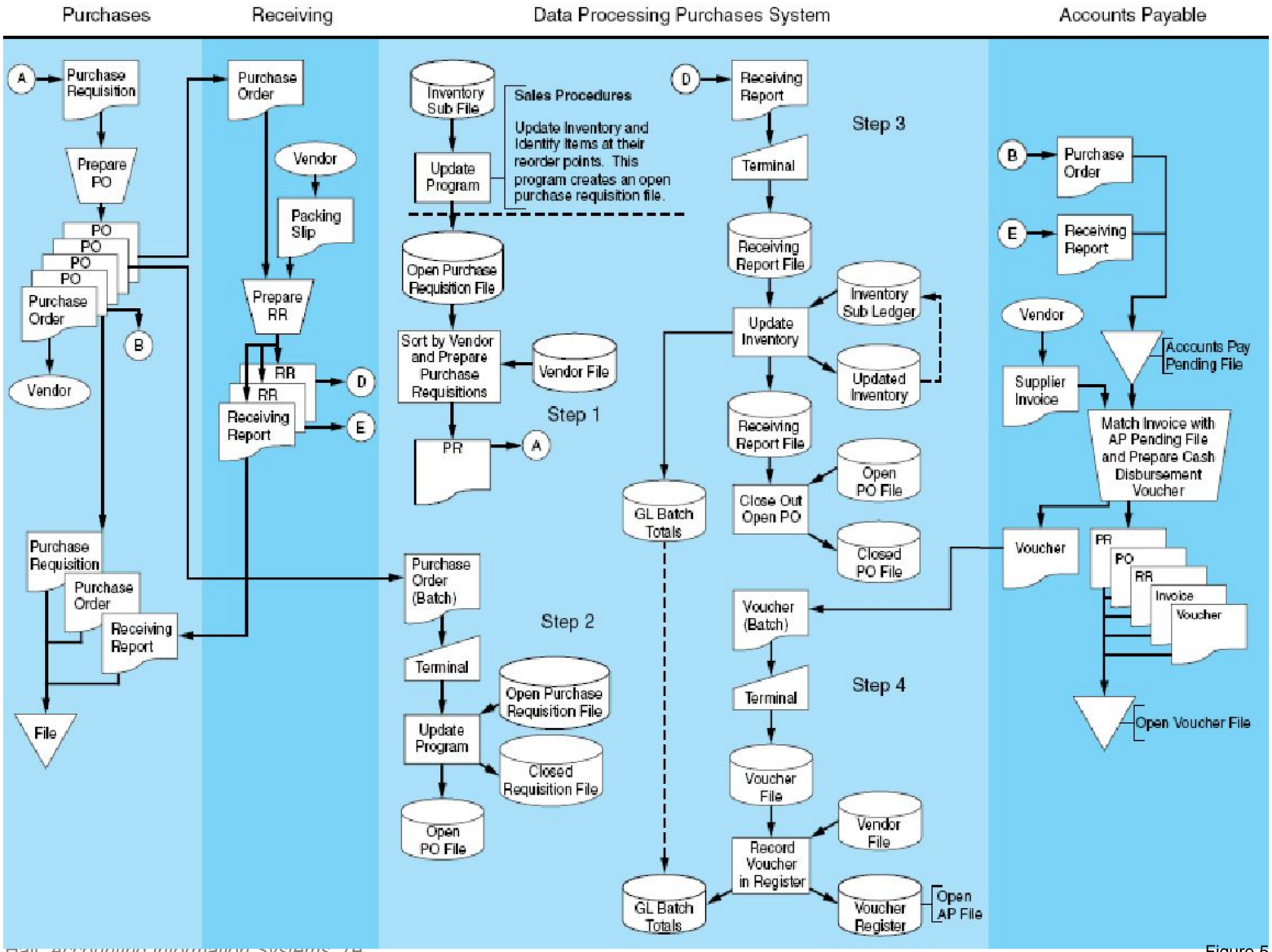
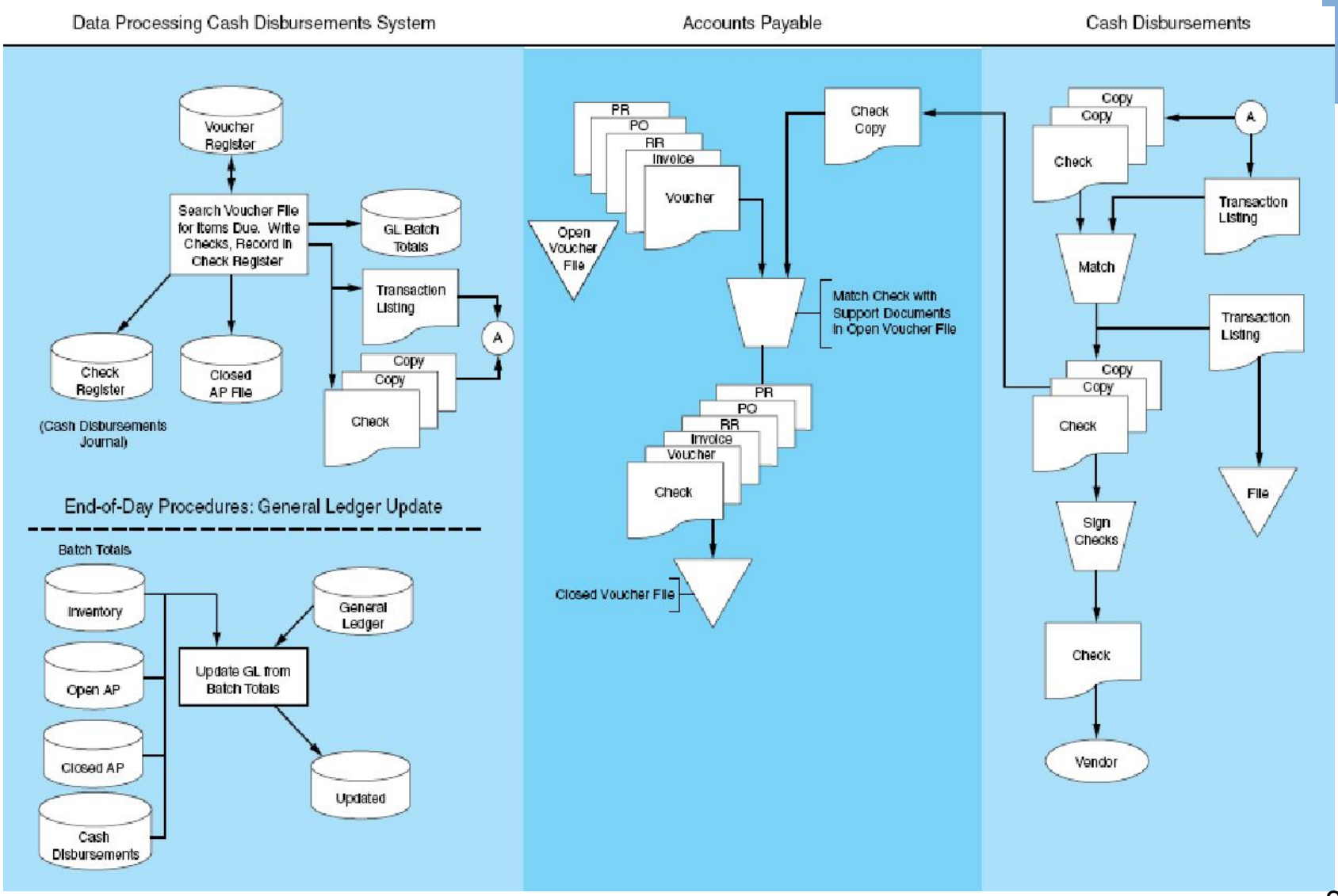


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Batch Purchases System *(continued)*



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Advantages of Real-Time Data Input & Processing Over Batch Processing

- Shortens the time-lag in record-keeping; hence, records are more current
- Eliminates much of the routine manual procedures, such as transcribing information onto paper documents
- Eliminates much of the storage and shuffling of paper documents
- Reduces data entry correction procedures

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Reengineered Purchases/ Cash Disbursements System

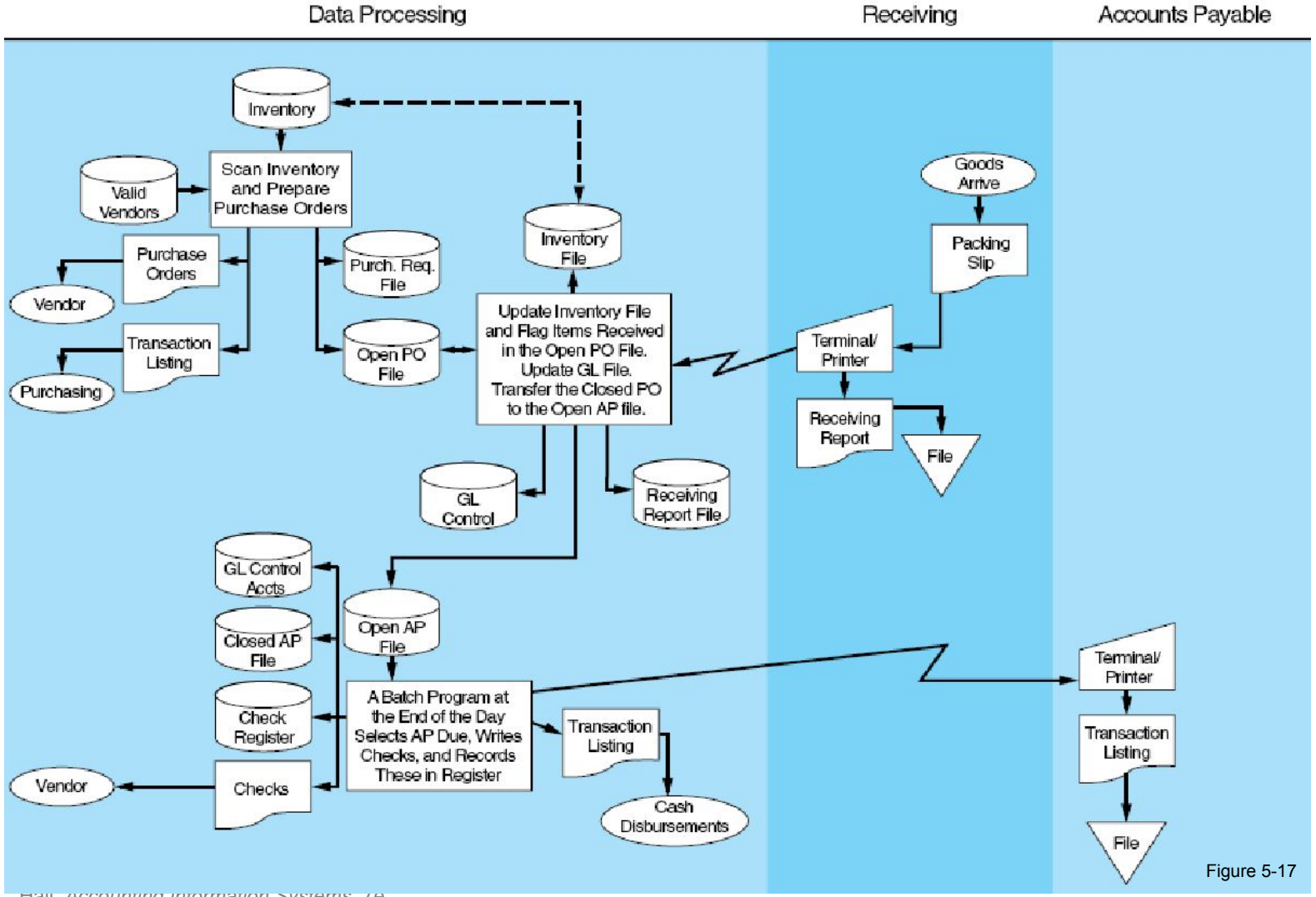


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Summary of Internal Controls

CONTROL POINTS IN THE EXPENDITURE CYCLE		
Control Activity	**Purchases Processing System**	**Cash Disbursements System**
Transactions authorization	Inventory control	Accounts payable authorizes payment.
Segregation of duties	Inventory control separate from purchasing and inventory custody. AP subsidiary ledger separate from the general ledger.	Separate AP subsidiary ledger, cash disbursements, and general ledger functions.
Supervision	Receiving department	
Accounting records	AP subsidiary ledger, general ledger, purchases requisition file, purchase order file, receiving report file.	Voucher payable file, AP subsidiary ledger, cash disbursements journal, general ledger cash accounts.
Access	Security of physical assets. Limit access to the accounting records above.	Proper security over cash. Limit access to the accounting records above.
Independent verification	Accounts payable reconciles source documents before liability is recorded. General ledger reconciles overall accuracy of process.	Final review by cash disbursements. Overall reconciliation by general ledger. Periodic bank reconciliation by controller.

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General Internal Controls

- Organization controls
  - segregation of duties
- Documentation
- Asset Accountability Controls
- Management Practices
- Data Center Operations Controls
- Authorization Controls
- Access Controls

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Authorization Controls

- Purchases of inventory should be authorized by the Inventory Control department, *not* by purchasing agents
- Accounts Payable authorizes the payments of bills, *not* the cash disbursements clerk, who writes the checks

How do these controls change in a CBAS?

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Computer-Based Authorization Controls

- Authorizations are automated.
  - programmed decision rules must be debugged
- Automating inventory in EDI and JIT
  - faulty inventory model can lead to over-purchasing or under-purchasing
- Cash disbursements may automate check printing and signing.
  - programming logic must be flawless
  - automated signing only below a dollar threshold

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Traditional Segregation of Duties

- Warehouse (stores)
- Inventory control
- Accounts payable
- General ledger
- Requisitioning
- Purchases
- Purchases returns and allowances
- Cash disbursements

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Segregation of Functions

- Custody of the asset, inventory, by the Warehouse must be separate from recordkeeping for the assets by the Inventory Control.
- Custody of the asset, cash, by Cash Disbursements must be kept separate from recordkeeping for the asset by A/P.

How do these controls change in a CBAS?

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Computer-Based Segregation of Functions

- Extensive consolidation by the computer of tasks traditionally segregated
  - computer programs authorize and process purchase orders
  - computer programs authorize and issue checks to vendors

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Supervision

- Within the expenditure cycle, **supervision** is of highest importance in the Receiving department, where the inventory arrives and is logged in by a receiving clerk. Need to minimize:
  - failures to properly inspect the assets
  - theft of the assets

How do these controls change in a CBAS?

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Computer-Based Supervision

- Automation often leads to a collapsing of the traditional segregation of duties.
  - requires greater supervision
- Supervision takes on new aspects as technology advances.
  - electronic monitoring
- Supervision because more difficult as the workplace becomes more sophisticated.
  - employees may have advanced IT training

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Accounting Records

- Must maintain adequate records for:
  - accounts payable
  - vouchers payable
  - checks
  - general ledger
  - subsidiary ledgers

How do these controls change in a CBAS?

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Computer-Based Accounting Records

- Accounting records rest on the reliability and security of stored digitalized data.
  - Accountants should be skeptical about the accuracy of hard-copy printouts.
  - Backups - the system needs to ensure that backups of all files are continuously kept
- Most automated systems still have a lot of paper documents.
  - This is good for audit trail purposes but is often inefficient.
  - As the system becomes increasing paperless, maintaining an audit trail becomes more difficult.

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Access Controls

- Access to:
  - inventories (direct)
  - cash (direct)
  - accounting records (indirect)

How do these controls change in a CBAS?

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Computer-Based Access Controls

- Magnetic records are vulnerable to both authorized and unauthorized exposure and should be protected
  - must have limited file accessibility
  - programs must be safeguarded and monitored

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Independent Verification

- A/Payable dept. verifies much of the work done within the expenditure cycle.
  - PR, PO, receiving reports, and suppliers' invoices must be checked and verified by A/P.
- G/Ledger dept. verifies:
  - the total obligations recorded equal the total inventories received
  - the total reductions in accounts payable equal the total disbursements of cash

How do these controls change in a CBAS?

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Computer-Based Independent Verification

- Automating the accounting function reduces the need for verification by reducing the chances of fraud and error in the expenditure cycle.
- However, the need for verification shifts to the computer program and the programmers where fraud and error may still be present.

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